

Final cloud computing project, a.y. 2025-26

1. Context

Modern cloud applications increasingly rely on large-scale distributed data processing pipelines exposed through service-oriented architectures. In many real-world scenarios, backend analytics systems based on Hadoop and Spark are integrated with REST services that allow users or external applications to submit jobs, monitor execution status, and retrieve computed results.

2. Goal of the project

The project must be developed by the same groups of three students that were used to carry out laboratories in class. Each group must design and implement a cloud-oriented distributed analytics system **including the following elements**:

- Hadoop MapReduce;
- Apache Spark;
- REST-based service interaction;
- containerized deployment.

3. Dataset selection

Students must autonomously identify and collect a dataset of at least:

- **1–1.5 GB** in size.

Example dataset categories include:

- textual datasets;
- system logs;
- web datasets;
- graph/network datasets;
- IoT datasets;
- multimedia metadata;
- mobility traces;
- open government datasets.

The dataset choice must be properly motivated in the final project report.

4. Use case and analytics workflow

Each group must autonomously define:

- the application scenario;
- the analytics objective;
- the distributed processing workflow.

Examples of possible analytics workflows include:

- multi-stage filtering and aggregation;
- ranking systems;
- recommendation pipelines;
- graph analytics;
- similarity analysis;
- trend detection;
- anomaly detection;
- distributed statistics extraction.

The application domain and analytics workflow are **intentionally left open**, allowing each group of students to design its own distributed data-processing use case.

The proposed workflow must be sufficiently non-trivial and appropriate for distributed execution.

5. Hadoop implementation

The Hadoop implementation must consist of **at least two non-iterative MapReduce jobs in cascade**, written in Java. To reach the maximum grade, students are **encouraged** to consider and properly motivate:

- combiners or in-mapper combining;
- custom partitioning strategies;
- setup() and cleanup() methods;
- custom writables;
- more than one reducer;
- performance comparison against a Python-based, non-parallel execution.

6. Spark implementation

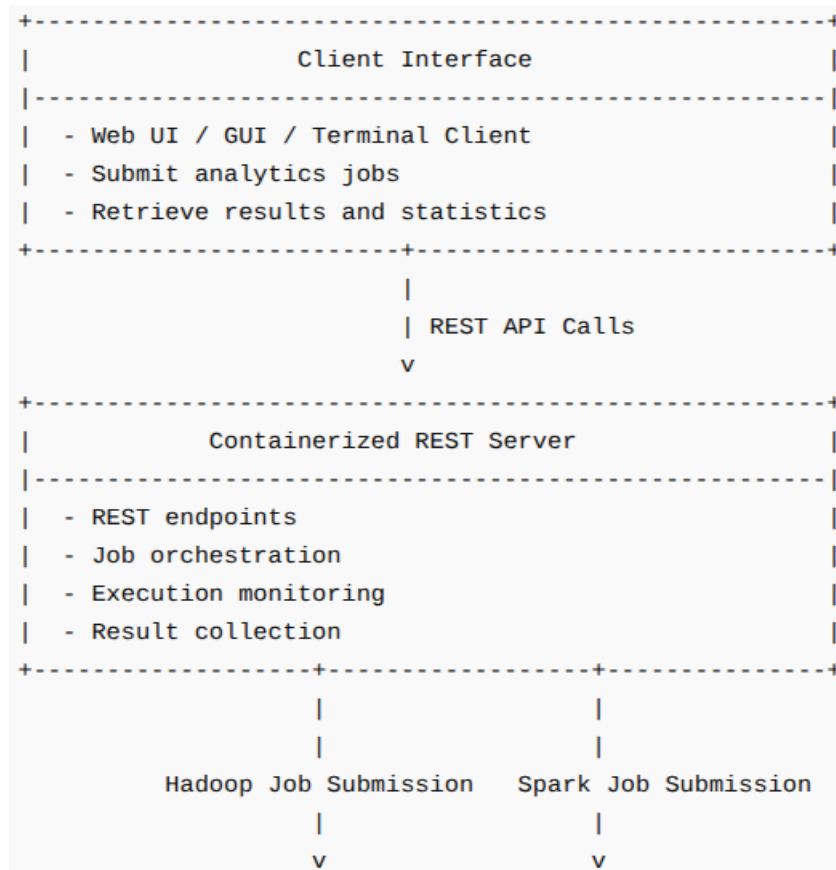
Students must implement an equivalent solution using Apache Spark, which:

- must be developed in Python;
- must reproduce the same analytics workflow implemented in Hadoop;
- must allow a meaningful comparison with the Hadoop solution and, optionally, with the Python non-parallel solution.

7. REST-based architecture

The project must expose the distributed analytics functionalities through a REST-oriented architecture. In particular, students must develop:

- a **REST client interface** (e.g., terminal-based, web-based, graphical). The interface must allow users to:
 - start Hadoop/Spark jobs;
 - select execution parameters;
 - retrieve results and execution statistics.
- a **containerized (Docker) REST server**. This server acts as an intermediate layer between the client interface and the distributed backend.



8. Experimental evaluation

Students must perform an experimental evaluation, which should compare Hadoop and Spark considering aspects such as:

- execution time;
- memory usage;
- CPU utilization;
- impact of configuration parameters (e.g., number of input splits, number of reducers);
- test with multiple dataset sizes.

9. Project tracking and final report

Students must create a private **GitHub** or **GitLab** repository, where they have to periodically commit project changes. The private repository must be shared only with the professors, using usernames carlo.vallati and carlo.puliafito (GitLab) or warner83 and cpuliafito (GitHub).

Students must also submit a brief **technical report** (recommended length: 4-5 pages) including:

- application scenario;
- dataset description;
- architecture description;
- MapReduce workflow pseudocode;
- REST architecture description;
- experimental evaluation;
- discussion of results.

FAQs on the Cloud computing project and final exam

1. When and how to upload the final project?

- The final project (code + report) must be uploaded through the **Google form** at: <https://forms.gle/ycfMLuy5qaMsgG1q8>;
- Each group has to submit their project as a .zip file named "Group{Name}_Session{DD-MM-YYYY}.zip";
- Please, do not include the input datasets in the .zip file;
- Project upload must be performed **three working days before** the official session day, at the latest. For example, if the official session is scheduled for the 10/06/2026 (Wednesday), students have time to upload their project by 04/06/2025 included (Thursday), hence allowing for three working days before the official session day (Friday, Monday, Tuesday). **Projects that are uploaded after the deadline will be considered for the next session;**
- Once the project is uploaded, **no further modifications** are allowed to it.

2. What about the project discussion?

- Project discussion will take place on the official session day or a few days before, depending on the number of projects submitted in that session. In the second case, I will inform you of the exact date and time;
- **All group members** must be present that day and discuss the project as part of their exam;
- Project discussion will consist of **two parts**:
 - For the first part (duration 15 minutes – hard time), students have to prepare a presentation (either through a separate PowerPoint or directly commenting the project report) of design choices, dataset and obtained results, and show an execution demo of their work on the cluster;
 - The second part (no duration pre-established) will be a detailed discussion of code.
- All the group members must show **complete knowledge** of each aspect of the project. If a student does not demonstrate active involvement in the different parts of the project, that student will have to develop the project assigned to single students and discuss it in another exam session;
- Discussion of the project is in English if at least one group member is a non-Italian speaker. Otherwise, the group can choose to discuss their project either in English or in Italian.

3. What about the oral exam?

- The overall exam consists of project discussion and the oral exams for Prof. Vallati's and Prof. Puliafito's parts;
- Students can take the different parts of the exam in **different sessions and in any order**;

- Each part of the exam will have a **validity of one year**. After that deadline, that part must be repeated by the student. For example, if you discuss one part on 10/06/2026 (first summer session), this will be valid until the first summer exam session of 2027 (**INCLUDED**);
- Oral exams will take place on official **session days** only;
- When you enroll for a session on the system, please **specify in the notes** which part(s) of the exam you are going to take in that session;
- **Groups do not stand** for oral exams: members of a given group can take oral exams in different sessions;
- Students can choose to take their oral exam either in English or in Italian.